

PART I - FOUNDATION PLC PROJECTS

Project 6 - Automated Bottle Filling System

Version 1 - Foundation PLC Project

PLC Platform: Allen-Bradley MicroLogix 1100

Programming Software: RSLogix Micro Starter Lite 8.30

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1. Project Overview

The Automated Bottle Filling System was developed to introduce a small automatic machine sequence using a maintained System Run state, bottle detection, timed filling, valve control, and conveyor movement.

Start Push Button I:1/0 latches internal bit B3:0/0 System_Run ON, while Stop Push Button I:1/1 unlatches it. When the system is running and a bottle is detected at I:1/2, the conveyor stops and Filling Valve O:2/1 opens. Timer T4:0 then controls a five-second filling period.

When the filling timer reaches its preset, the valve closes and Conveyor Motor O:2/0 restarts so the filled bottle can leave the station. Once the bottle clears the sensor, the non-retentive timer resets and the conveyor continues waiting for the next bottle.

This project was an important progression from individual timer, counter, and motor-control exercises into a coordinated automatic sequence.

2. Engineering Objective

The objective of this project was to develop and verify a PLC program that automatically moved a bottle into a filling position, stopped the conveyor, filled the bottle for five seconds, closed the filling valve, and restarted the conveyor.

- Latch a System Run condition from the Start command.
- Remove the System Run condition from the Stop command.
- Run the conveyor while no bottle is detected.
- Stop the conveyor when a bottle reaches the filling position.
- Open the filling valve while the bottle is present and the fill timer is not complete.
- Time the filling process for five seconds.
- Close the valve when filling is complete.
- Restart the conveyor automatically after filling.
- Reset the fill timer when the bottle leaves the sensor.

3. Functional Requirements

- Start Push Button I:1/0 shall latch System_Run B3:0/0 ON.
- Stop Push Button I:1/1 shall unlatch System_Run B3:0/0.
- Conveyor Motor O:2/0 shall operate when System_Run is true and no bottle is detected.
- Bottle Sensor I:1/2 shall identify the bottle at the filling position.
- When System_Run and Bottle Sensor are both true, Timer T4:0 shall begin timing.
- T4:0 shall use a 0.01-second time base and a preset of 500, producing a five-second fill period.
- Filling Valve O:2/1 shall energize when System_Run is true, Bottle Sensor is true, and T4:0/DN is false.
- Filling Valve O:2/1 shall de-energize when T4:0/DN becomes true.
- Conveyor Motor O:2/0 shall restart after T4:0/DN becomes true even while the bottle is still detected.
- When the bottle leaves the sensor, the timer rung shall become false and T4:0 shall reset.
- Activating Stop shall remove System_Run and de-energize both machine outputs through the dependent control logic.

4. System Components

Component	Identification	Function
PLC Platform	Allen-Bradley MicroLogix 1100	Executes the bottle-filling sequence
Input Module	1762-IQ16	Provides Start, Stop, and bottle-sensor inputs
Output Module	1762-OB16	Provides conveyor and filling-valve outputs
Start Push Button	I:1/0	Latches the System Run condition
Stop Push Button	I:1/1	Clears the System Run condition
Bottle Sensor	I:1/2	Detects a bottle at the filling position
System Run	B3:0/0	Maintains the machine operating state
Filling Timer	T4:0	Provides the five-second fill period
Conveyor Motor	O:2/0	Moves bottles through the station
Filling Valve	O:2/1	Controls product flow into the bottle
Programming Software	RSLogix Micro Starter Lite 8.30	Ladder logic development and monitoring

5. I/O Assignment

Physical I/O

Device	PLC Address	Type	Description
Start Push Button	I:1/0	Digital Input	Starts/latches the system
Stop Push Button	I:1/1	Digital Input	Stops/unlatches the system
Bottle Sensor	I:1/2	Digital Input	Detects bottle at filling station
Conveyor Motor	O:2/0	Digital Output	Conveyor command
Filling Valve	O:2/1	Digital Output	Filling command

Internal PLC Data

Function	Address	Type	Purpose
System Run	B3:0/0	Internal Bit	Maintains machine run state
Filling Timer	T4:0	TON	Times the filling operation
Timer Done	T4:0/DN	Status Bit	Indicates fill period completion
Timer Time Base	-	0.01 second	Timer resolution
Timer Preset	-	500	Five-second fill period

Fill time: $500 \times 0.01 \text{ second} = 5.00 \text{ seconds}$.

6. Control Strategy

I used an internal System_Run bit to separate the machine operating state from the physical outputs. Start latches B3:0/0 and Stop unlatches it.

While the system is running and no bottle is detected, the conveyor is energized. When a bottle reaches the sensor, the normal conveyor branch becomes false, stopping the bottle at the filling position.

At the same time, System_Run and Bottle_Sensor enable T4:0 and energize the Filling Valve while T4:0/DN remains false. The bottle therefore fills during the timer period.

Once the timer completes, the valve closes and the timer-done branch in the conveyor rung becomes true, restarting the conveyor. When the bottle leaves the sensor, T4:0 resets and the conveyor continues through the no-bottle branch.

7. Engineering Design Decisions

Design Decision	Reason
Used internal System_Run B3:0/0	Separated maintained machine state from physical outputs
Used OTL/OTU on System_Run	Allowed momentary Start and Stop commands to maintain/clear the run state
Used Bottle Sensor to stop conveyor	Positioned the bottle for filling
Used TON T4:0	Provided a repeatable five-second fill duration
Used XIO T4:0/DN in valve rung	Kept the valve open only until filling completed
Used timer-done branch in conveyor rung	Restarted the conveyor automatically after filling
Allowed timer to reset when bottle leaves	Prepared the sequence for the next bottle without a separate reset command
Kept Version 1 without separate cycle-state bits	Maintained a compact foundation sequence before adding industrial diagnostics

A key limitation is that bottle presence, valve operation, and conveyor motion are assumed from commanded states. The program does not verify physical feedback, detect a stuck bottle, or latch a filling fault.

8. Ladder Logic Description

The Version 1 program contains five functional rungs. The clean representations below follow the approved single-line portfolio format.

8.1 Rung 0000 - Start / Latch System Run

```
|----[ XIC I:1/0 ]------( OTL B3:0/0 )----|
      Start_PB                System_Run
```

When Start Push Button I:1/0 becomes true, B3:0/0 System_Run is latched ON and remains true after the button is released.

8.2 Rung 0001 - Stop / Unlatch System Run

```
|----[ XIC I:1/1 ]------( OTU B3:0/0 )----|
      Stop_PB                System_Run
```

When Stop Push Button I:1/1 becomes true, B3:0/0 is unlatched. The remaining control rungs depend on System_Run, so the conveyor and valve commands are removed.

8.3 Rung 0002 - Bottle Filling Timer

```
|----[ XIC B3:0/0 ]----[ XIC I:1/2 ]-----[ TON T4:0 ]----|
      System_Run      Bottle_Sensor      Filling_Timer  PRE: 500   Time Base: 0.01
```

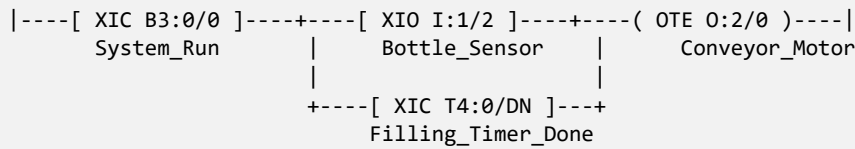
When the system is running and a bottle is detected, T4:0 begins timing. If the bottle sensor clears before five seconds, the non-retentive timer resets.

8.4 Rung 0003 - Filling Valve Control

```
|----[ XIC B3:0/0 ]----[ XIC I:1/2 ]----[ XIO T4:0/DN ]----( OTE O:2/1 )----|
      System_Run      Bottle_Sensor      Filling_Timer_Done  Filling_Valve
```

The Filling Valve is energized while the system is running, the bottle is present, and the filling timer is not complete. When T4:0/DN becomes true, the XIO condition opens logically and the valve turns OFF.

8.5 Rung 0004 - Conveyor Motor Control



The conveyor operates when System_Run is true and either no bottle is detected or the filling timer has completed. A detected bottle stops the conveyor during filling, and T4:0/DN restarts the conveyor at the end of the fill period.

8.6 Actual RSLogix Implementation

The figure below shows the corresponding ladder logic as implemented in RSLogix Micro Starter Lite.

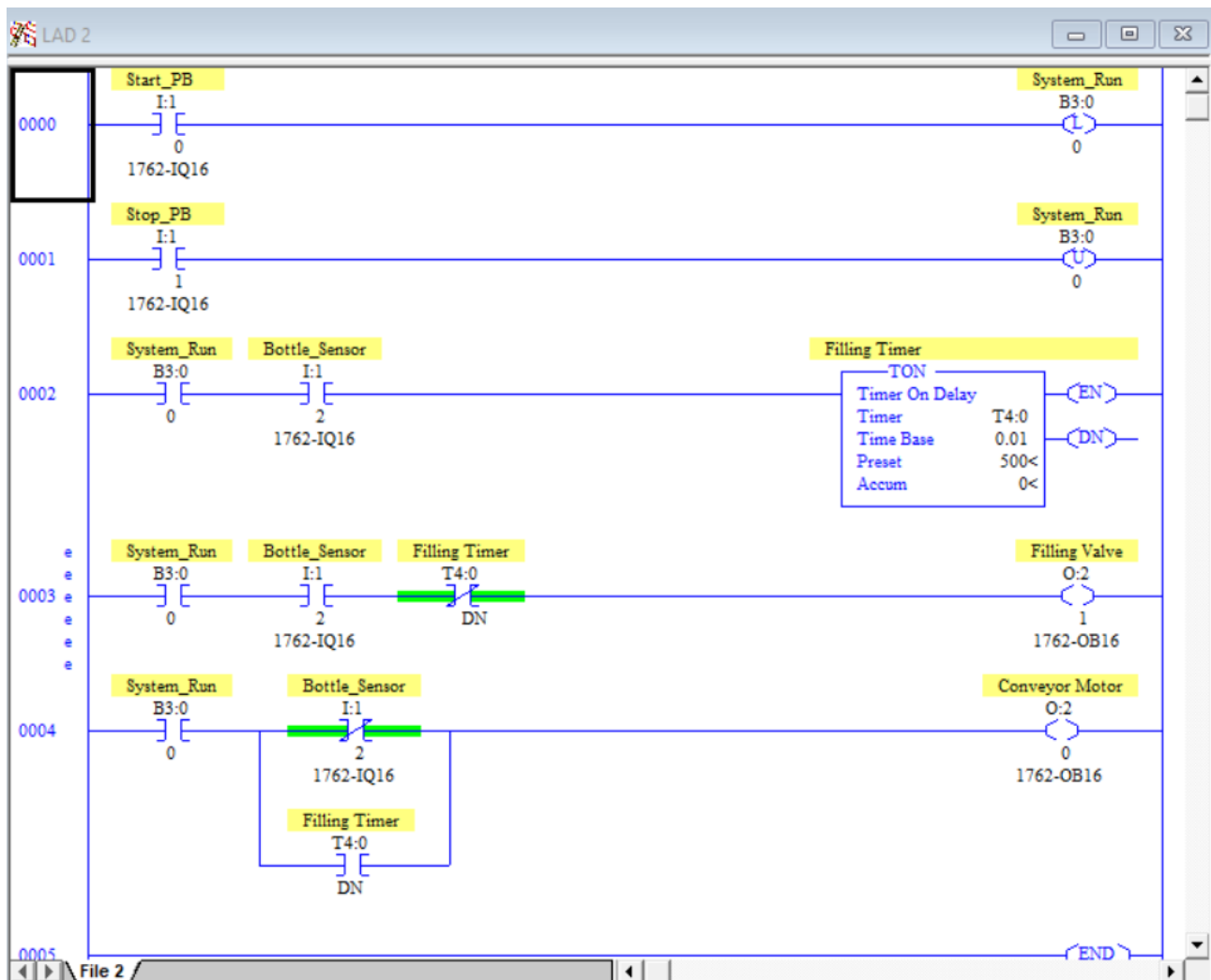


Figure P6.1 - RSLogix Implementation of the Version 1 Automated Bottle Filling System

9. Sequence of Operation

Step 1 - Start Command: The operator presses Start Push Button I:1/0.

Step 2 - System Run Latched: B3:0/0 becomes true.

Step 3 - Conveyor Runs: With no bottle detected, Conveyor Motor O:2/0 is energized.

Step 4 - Bottle Arrives: Bottle Sensor I:1/2 becomes true.

Step 5 - Conveyor Stops: The no-bottle branch becomes false and the bottle remains at the filling position.

Step 6 - Filling Begins: T4:0 begins timing and Filling Valve O:2/1 energizes.

Step 7 - Five-Second Fill Completes: T4:0/DN becomes true.

Step 8 - Filling Valve Closes: XIO T4:0/DN becomes false, removing O:2/1.

Step 9 - Conveyor Restarts: The T4:0/DN branch energizes Conveyor Motor O:2/0.

Step 10 - Bottle Leaves Sensor: I:1/2 returns false.

Step 11 - Timer Resets: T4:0 resets and the conveyor continues through the no-bottle branch.

Step 12 - Next Cycle: The system remains ready for the next bottle until Stop is activated.

10. Testing & Validation

The final ladder diagram represents the tested Version 1 implementation. I documented the functional tests that directly verify the actual sequence and marked them PASS.

Test ID	Test Action	Expected Result	Observed Result	Status
T-01	Press Start Push Button	System_Run latches ON	System_Run latched ON	PASS
T-02	System running with no bottle present	Conveyor Motor ON	Conveyor ran	PASS
T-03	Bottle Sensor becomes true	Conveyor stops; Filling Valve opens; timer starts	Conveyor stopped, valve opened, timer started	PASS
T-04	Bottle remains present for less than 5 seconds	Valve remains ON; conveyor remains stopped	Valve stayed ON; conveyor stayed stopped	PASS
T-05	Filling timer reaches 5 seconds	Valve closes; conveyor restarts	Valve closed and conveyor restarted	PASS
T-06	Bottle leaves sensor after filling	Timer resets; conveyor continues	Timer reset and conveyor continued	PASS
T-07	Press Stop during normal conveyor operation	System_Run clears; conveyor stops	System_Run cleared and conveyor stopped	PASS
T-08	Press Stop during filling	System_Run clears; valve OFF; timer resets; conveyor OFF	Outputs removed and timer reset	PASS
T-09	Bottle Sensor clears before timer completes	Timer resets; valve OFF; conveyor resumes	Timer reset, valve closed, conveyor resumed	PASS

Validation Summary

Testing confirmed the complete bottle-filling sequence from machine start through bottle detection, timed filling, conveyor restart, and timer reset. The Stop command also correctly removed System_Run during both normal conveyor operation and an active fill.

All nine defined Version 1 functional tests passed. The testing represents functional PLC program validation within the project development environment, not industrial commissioning or safety validation.

11. Results

The completed Automated Bottle Filling System successfully coordinated conveyor movement and timed filling using the bottle sensor and a five-second TON timer.

The conveyor ran while no bottle was present, stopped when a bottle reached the filling position, and remained stopped while the Filling Valve was energized. When the filling timer completed, the valve closed and the conveyor restarted automatically.

The timer reset when the bottle left the sensor, allowing the sequence to repeat. All nine defined functional tests passed.

12. Challenges & Troubleshooting

A useful troubleshooting path for this project is: System_Run -> Bottle Sensor -> Filling Timer -> Filling Valve / Conveyor Branch Logic -> Bottle Leaves -> Timer Reset.

Symptom	Possible Cause	Diagnostic Check	Corrective Action
System will not run	Start input or System_Run latch not true	Monitor I:1/0 and B3:0/0	Verify Start rung and OTL B3:0/0
System will not stop	Stop input or unlatch rung not true	Monitor I:1/1 and B3:0/0	Verify Stop rung and OTU B3:0/0
Conveyor does not stop at bottle	Bottle Sensor not true or rung branch incorrect	Monitor I:1/2 and O:2/0	Verify Bottle Sensor and conveyor branch
Filling Valve does not open	System_Run/Bottle Sensor condition false or timer already done	Monitor B3:0/0, I:1/2, T4:0/DN	Verify valve rung conditions
Valve stays ON too long	Timer not reaching preset or Done bit incorrect	Monitor T4:0.ACC and T4:0/DN	Verify timer configuration
Conveyor does not restart after fill	Timer Done branch not true	Monitor T4:0/DN and Rung 0004	Verify timer completion and branch
Timer does not reset after bottle leaves	Bottle Sensor remains true	Monitor I:1/2 and T4:0	Verify sensor clears
Fill aborts early	Bottle Sensor clears before 5 seconds	Monitor I:1/2 during fill	Verify bottle positioning/sensor behavior

The project reinforced that automatic sequencing often depends on the relationship between multiple rung conditions rather than a single output instruction.

13. Lessons Learned

This project helped me move from individual control functions into a complete automatic sequence.

I learned how an internal System_Run bit can represent overall machine state while physical outputs are controlled separately.

The project showed how a timer Done bit can be used not only to end one action, such as filling, but also to initiate the next action, such as conveyor restart.

I also learned that the same bottle sensor participates in several parts of the sequence: it stops the conveyor, enables filling, and resets the cycle when the bottle leaves.

Testing Stop behavior during an active fill reinforced the importance of ensuring that a machine-wide run condition is included in each relevant output path.

14. Industrial Relevance

Timed bottle-filling sequences are common in packaging, beverage, chemical, pharmaceutical, and general process-manufacturing applications.

The Version 1 project demonstrates the foundation of an indexing/filling sequence in which a conveyor positions a container, a valve operates for a defined period, and the conveyor then advances the completed container.

A real industrial filling system would normally require additional features such as valve feedback, container-position verification, no-bottle/no-fill logic, conveyor feedback, jam detection, product level or flow measurement, batch counting, alarms, operator modes, safety interlocks, fault recovery, and HMI status.

This Version 1 project therefore represents a foundation automatic filling sequence rather than a production-ready packaging machine.

15. Transition to Industrial Version (Version 2)

Version 2 can retain the same basic sequence while introducing explicit machine states, diagnostics, fault handling, operator modes, and stronger control separation.

Version 2 Improvement	Engineering Benefit
Explicit Fill Cycle Active state	Makes the machine sequence easier to monitor and troubleshoot
Auto/Manual operating modes	Supports production and maintenance operation
Mode-conflict detection	Identifies invalid simultaneous mode commands
Conveyor permissives/interlocks	Prevents motion under invalid conditions
Valve permissive and feedback	Verifies filling command and device response
Bottle-present verification	Improves confidence that a container is correctly positioned
No-bottle/no-fill logic	Prevents product discharge without a container
Bottle-stuck / sensor-stuck diagnostics	Detects abnormal sensor conditions
Fill timeout fault	Detects a cycle that does not complete as expected
Jam detection	Detects product-flow problems
Batch counter	Tracks completed bottles
Fault latch and controlled reset	Provides defined abnormal-condition recovery
HMI machine status	Displays mode, run state, bottle status, timer, valve, conveyor, and faults
Adjustable fill time with limits	Allows authorized process adjustment
Production statistics / SCADA integration	Supports higher-level monitoring and reporting

A stronger Version 2 architecture could become: Start Request -> System Run -> Mode / Permissive Checks -> Conveyor Index -> Bottle Verify -> Fill Cycle Active -> Timed Fill -> Fill Complete -> Conveyor Release -> Cycle Complete -> Diagnostics / HMI.

The improvements should preserve the simplicity of the original sequence while making machine state, fault conditions, and recovery behavior explicit.

16. Project Summary

The Automated Bottle Filling System coordinated a conveyor, bottle sensor, filling valve, and five-second timer in a repeatable automatic sequence.

Start and Stop commands controlled internal System_Run B3:0/0. While the system was running, the conveyor moved until Bottle Sensor I:1/2 detected a bottle. The conveyor then stopped, Filling Valve O:2/1 opened, and T4:0 timed the filling period.

When the timer completed, the valve closed and Conveyor Motor O:2/0 restarted. When the bottle left the sensor, the timer reset and the system became ready for the next bottle.

All nine defined functional tests passed. This project strengthened my understanding of maintained machine state, timed sequencing, branch logic, automatic cycle progression, and stop behavior, and it provides a strong foundation for the more structured industrial Version 2 filling system.